

Burak Kadron

Software Engineer, Security Tooling | PhD in Computer Science

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Summary

Software engineer and security researcher with a PhD in program analysis and 4 years building fuzzing and constraint-solving tools for smart-contract security. Experienced in language design (parsers, interpreters, DSLs), input generation, and side-channel analysis, with publications at NDSS, ESEC/FSE, ISSTA, and ASE.

Experience

Veridise, Research Scientist

Remote

Nov 2022 – present

- Lead engineer on OrCa, Veridise's fuzzer for Solidity smart contracts (Python), used across the company's security audits
- Rewrote the LTL-based [V] specification language end to end: updated ANTLR grammar, a generator that converts parse trees to an intermediate representation, and the interpreter
- Designed and built from scratch a hint language that lets users specify how functions are fuzzed; user constraints are compiled to SMT-LIB and solved with Z3 to generate function inputs
- Added on-chain fuzzing support, enabling OrCa to test deployed contracts against live chain state
- Maintained OrCa's Rust EVM component, keeping it in sync with upstream dependencies including Foundry's EVM
- Refactored core fuzzing components, improving performance and coverage
- Performed smart-contract and infrastructure security audits for 3 client protocols: ThetaNuts, Manta Network, and Stellar (Soroban core)

University of California, Santa Barbara, Graduate Student Researcher

Santa Barbara, CA

Jan 2017 – Sept 2022

- Developed techniques to detect, quantify, and mitigate side-channel information leakage in encrypted network traffic
- Built [TSA](#), an open-source tool for network side-channel detection and quantification
- Published at NDSS, ISSTA, and ESEC/FSE; represented UCSB Verification Lab in DARPA's STAC (Space/Time Analysis for Cybersecurity) program
- Mentored one Master's and two undergraduate students on research projects, including CorbFuzz, a fuzzer for checking browser security policies (ASE 2021)

NASA Ames Research Center (KBR Inc.), Research Intern

Remote

June 2020 – Sept 2020

- Analyzed robustness of Boeing's runway-image neural network for autonomous center-line tracking
- Built a concolic analysis method that uses the Marabou verifier to check robustness to input perturbations and generate counterexample images
- Published findings as a case study (VSTTE 2021)

Carnegie Mellon University, CyLab, Research Intern

Moffett Field, CA

June 2018 – Sept 2018

- Developed input generation techniques for side-channel analysis; represented CMU CyLab in DARPA STAC
- Worked on adversarial input generation for neural networks using symbolic execution (ISSRE 2019)

University of California, Santa Barbara, Teaching Assistant
• Data Structures & Algorithms I & II, Formal Languages & Automata

Santa Barbara, CA
Sept 2016 – June 2018

Airties Wireless Networks, Systems & Research Engineer (Part-time)
• Implemented mesh-networking code in C supporting client handover between Wi-Fi routers

Istanbul, Türkiye
June 2013 – Sept 2015

Education

PhD **University of California, Santa Barbara**, Computer Science
• Dissertation: Detection, Quantification and Mitigation of Network Side Channels
• Advisor: Tefvik Bultan
• Committee: Tefvik Bultan, Giovanni Vigna, Yu-Xiang Wang

Santa Barbara, CA
Sept 2016 – Sept 2022

BS **Boğaziçi University**, Computer Engineering

Istanbul, Türkiye
Sept 2010 – June 2014

Skills

Languages: Python, Rust, Solidity, Java, C, Bash

Security & Analysis: Fuzzing, smart-contract auditing, side-channel analysis, SMT-based constraint solving, concolic/symbolic execution

Tools: Z3, CVC5, ANTLR, Foundry, EVM, Docker, Git

Selected Publications

Fuzzing, Symbolic Execution, and Expert Guidance for Better Testing

Jan 2023

Burak Kadron, Yannic Noller, Rohan Padhye, Tefvik Bultan, Corina S. Păsăreanu, Koushik Sen
(IEEE Software)

TSA: A Tool to Detect and Quantify Network Side-Channels

Nov 2022

Burak Kadron, Tefvik Bultan
(ESEC/FSE 2022)

CorbFuzz: Checking Browser Security Policies with Fuzzing

Nov 2021

Chaofan Shou, **Burak Kadron**, Qi Su, Tefvik Bultan
(ASE 2021)

Feedback-Driven Side-Channel Analysis for Networked Applications

July 2020

Burak Kadron, Nicolás Rosner, Tefvik Bultan
(ISSTA 2020)

Profit: Detecting and Quantifying Side Channels in Networked Applications

Feb 2019

Nicolás Rosner, **Burak Kadron**, Lucas Bang, Tefvik Bultan
(NDSS 2019)

Service

Reviewer

2026

SPLASH/ISSTA 2026 Poster Session

Reviewer

2024

ACM Transactions on Software Engineering and Methodology (TOSEM)

President, Graduate Student Executive Committee

2022

UCSB CS Department

Treasurer, Graduate Student Executive Committee

2020

UCSB CS Department

Local Arrangements and Finance Chair

2020

ISSTA 2020